
SKILLS/ACCOLADES

- **Programming Languages:** Python, C++, C, JavaScript/TypeScript, SQL, Neo4j Cypher, Bash
- **Frameworks & Agents:** PyTorch, HuggingFace, Google ADK, LangGraph, LangChain, LlamaIndex, DSPy, PydanticAI, Temporal, vLLM, LLaMA-Factory
- **AI/ML:** Deep Learning, NLP, Vision-Language Models, LoRA Fine-tuning, Set-of-Mark Grounding, Speculative Decoding (EAGLE-3), Retrieval-Augmented Generation, Reciprocal Rank Fusion, Graph ML, Knowledge Graphs
- **Infra & Data:** AWS (EKS/S3/KMS), Azure, GCP, Kubernetes, Docker, Flux (GitOps), SOPS, Neo4j, Turbopuffer, OpenSearch, Redis, PostgreSQL, MongoDB, Git
- **Accolades:** Awarded the India-AI Fellowship by the Govt. of India as part of its IndiaAI Mission to foster AI research.

EXPERIENCE

- **Landeed** Hyderabad, India
Foundational Software and AI Engineer *June 2025 – Present*
AITSR — AI Title Search Report
 - Architected a hybrid LangGraph + Google ADK multi-agent pipeline that automates lawyer-grade land-title risk reports for Indian real estate, replacing manual attorney diligence for B2B lenders (banks and housing-finance NBFCs).
 - Designed a layered `default` → `state` → `client` analysis-profile framework with composable markdown “skill-packs” (99 skill-packs, 17 report formats, 32 deed types, 78 document types) behind a `ContextProvider` seam for drop-in RAG — replacing a hardcoded per-client prompt monolith.
 - Built a deterministic citation verifier gating $\geq 98\%$ first-pass citation accuracy (every claim traceable to a source page, no LLM-as-judge) over a ~ 50 -report eval set, plus a 49-card statute-cited legal knowledge base (41 subjects, 38 critical) with a lawyer-review feedback loop.
 - Productionized an end-to-end deed → EC → report flow as a durable Temporal workflow with an autonomous agent fetching Encumbrance Certificates from state government portals (TS/AP/TN/KA), SHA-256 EC-reuse caching and per-stage metrics — eliminating orphaned-report failures from prior in-process background tasks.**Terra — Conversational land-records assistant**
 - Built Terra on Google ADK over AWS EKS: a root agent plus 4 specialized human-in-the-loop sub-agents orchestrating ~ 100 tools across 9 Indian states, unifying fragmented government land records behind one chat interface (AG-UI/CopilotKit streaming).
 - Shipped a Gemini + PydanticAI structured-extraction microservice hitting 100% success and field completeness on benchmark and $\sim 5\times$ faster than free-form (9.6s vs 49s/doc); load-tested to 100 concurrent users with HPA autoscaling and Redis-Streams SSE resilient to 20–40s portal fetches, shipped via Flux GitOps (SOPS/KMS).**Knowledge Graph**
 - Engineered a Neo4j Encumbrance-Certificate transaction knowledge graph with hybrid vector + full-text retrieval fused via Reciprocal Rank Fusion, plus a Turbopuffer batch index (Voyage `voyage-4-large` embeddings) with multilingual-safe FTS for Kannada/Telugu names.**Deed Intelligence — Grounded VLM fine-tuning**
 - Fine-tuned dots.mocr (3B document VLM) with a Set-of-Mark grounding protocol (LoRA over both vision and language towers, LLaMA-Factory) for verifiable key-information extraction from registered Indian deeds — lifting grounded field F1 from 0.04 to 0.73 and evidence-to-party linking from 0% to 91.8% with zero JSON parse failures, beating zero-shot Gemini 3.5 Flash on every metric.
 - Trained an EAGLE-3 speculative-decoding draft delivering a $2.27\times$ wall-clock speedup at token-exact parity (1.19s vs 2.71s/page), keeping a local 3B model competitive with frontier APIs at zero per-call cost and on-premise data residency.
- **ABB Ability Innovation Centre** Hyderabad, India
Data Science Research Intern *May – Aug 2024*

- **AI-based Question Answering:** Developed a novel state-of-the-art Retrieval-Augmented Generation (RAG) pipeline answering single- and multi-hop queries over an unstructured knowledge base, integrating Vector RAG and Graph RAG with a PDF-parsing and information-processing suite.
- **Ontology Creation:** Developed a standardized ontology-creation methodology from the pipeline's knowledge graph, enabling interoperability across correlated domains and consolidation of the Q&A system.

- **Mindcase Technologies** New Delhi, India
Machine Learning Engineer/Researcher *Mar – May 2024*
 - **Cloud ChatBot System:** Built a cloud-based Q&A chat application ensuring secure data flow between LLM deployment, SQL database and vector index; strengthened the Q&A pipeline with novel RAG methods robust across query types and benchmarks.
 - **Speech Recognition/Transcription:** Integrated AWS speech recognition, transcription and Textract via WebSocket APIs to enable live language recognition and OCR use cases within applications.

PROJECTS

- **ML/DL Security:** Investigated secure memory management, cryptographic protocols and access-control mechanisms to safeguard ML workloads on GPUs using hardware-based security features like TEEs; reproduced covert and side-channel attacks on multi-GPUs.
- **LLM-Generated Text Detection:** Built an LLM pipeline to detect AI-generated text: a SentencePiece + Byte-Pair-Encoding and TF-IDF ensemble (CatBoost, LightGBM, Naive Bayes) alongside a LoRA-finetuned DistilBERT; the ensemble scored 0.96 on the leaderboard, outperforming the standalone DistilBERT (0.86).

PUBLICATIONS

- ***OntoRAG*:** An Automated RAG Pipeline to derive Ontology for an Unstructured Knowledge Base *2024*
Yash Tiwari, Owais Ahmad Lone — unpublished manuscript (prepared; not submitted for publication).

EDUCATION

- **Indian Institute of Technology** Kharagpur, India
Bachelor of Technology in Computer Science and Engineering; GPA: 8.55/10 *Nov. 2021 – Apr. 2025*
 - **Relevant Courses:** Cryptography and Network Security, Usable Security and Privacy, Machine Learning, Deep Learning, Interpretable Machine Learning.